

KijaniKiosk DevOps Capstone

Final Scope - Track A: Infrastructure-First Delivery

Final implementation: kk-payments v1.2.0

Purpose

Deliver a controlled, observable and recoverable DevOps workflow for a containerized payments service, with staging-first promotion, infrastructure as code, security and monitoring gates, and a staging-only AWS receipt chain.

In Scope

- Terraform provisioning of the isolated kijani-staging Kubernetes namespace.
- Ansible management and validation of staging environment configuration.
- Jenkins CI/CD from master: CI, infrastructure validation, staging deploy, smoke test, error-rate gate, manual production approval, and production deploy.
- Same kk-payments Kubernetes Deployment manifest and v1.2.0 image in staging and production, with environment-specific ConfigMaps and Secrets.
- Log-based monitoring: 20 health probes; error rate greater than 5% blocks production.
- AWS staging receipt chain: S3 incoming -> Receipt Processor Lambda -> processed -> Notifier Lambda -> notifications.
- Security controls including npm audit, least-privilege IAM, Kubernetes Secrets, private encrypted/versioned S3, and generated-artifact exclusions.

Acceptance Evidence

- Jenkins master build completed successfully after staging gates and manual approval.
- Production: kk-payments v1.2.0 with 3 available replicas.
- Monitoring: 20/20 HTTP 200 responses; final error rate 0.00%.

Out of Scope / Known Limitations

Receipt publishing is intentionally staging-only; the notifier records a SENT notification in S3 rather than sending SMS/email; the local Minikube workload uses a Kubernetes Secret for the staging AWS writer key; serverless Terraform state is local and excluded from Git; the capstone primarily runs on local Minikube and Jenkins rather than a managed Kubernetes platform.

Delivery and Environment Model

Staging - kijani-staging

NODE_ENV=staging
DB_HOST=kk-postgres-staging
AWS_REGION=eu-north-1
RECEIPTS_BUCKET=kk-payments-receipts-staging

Production - kijani-project

NODE_ENV=production | DB_HOST=kk-postgres
RECEIPTS_BUCKET unset - receipt publishing disabled

Promotion Controls

- CI build/tests and npm security audit must pass.
- Terraform and Ansible validation must pass.
- Staging rollout and /health smoke test must succeed.
- Staging error-rate check must remain within threshold.
- A human must approve production promotion.
- Production prerequisites are verified before deployment.

- Security: final npm audit reported 0 vulnerabilities.
- Real staging receipt verified across incoming, processed and notifications S3 objects.
- Processed status = PROCESSED; notification status = SENT.